

Liang (Twist) Shan



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🏠 <https://twist-shan.github.io/>

RESEARCH INTEREST



- Statistical Foundations and Optimization Algorithms for **Modern Machine Learning**, especially in Deep Learning (DL), Reinforcement Learning (RL), Large Language Models (LLMs), and Diffusion Language Models (DLMs).

EDUCATION



Peking University

Beijing, China

- *B.S. in Mathematics, School of Mathematical Sciences* Jun. 2024–Jul. 2028 (Expected)
Second Major B.A. in Economics, National School of Development Sep. 2025–Jul. 2028 (Expected)
 - **Mathematics Courses:** Mathematical Analysis, Advanced Algebra, Geometry, Probability Theory, Complex Analysis, Ordinary Differential Equations, Mathematical Statistics, Applied Stochastic Processes, Statistical Thinking, Stochastic Analysis and Applications.
 - **Computer Science Courses:** Data Structure and Algorithm, Machine Learning and Artificial Intelligence, Fundamentals of Artificial Intelligence.
 - **Finance and Economics Courses:** Intermediate Microeconomics, Intermediate Macroeconomics, Econometrics(Honor), Mathematical Economics, Introduction to Finance, Security Investment, Mathematical Methods in Finance, Modern Quantitative Trading Systems, Topics in Quantitative Finance.

University of Copenhagen

Copenhagen, Denmark

- *International Exchange Student, Department of SCIENCE* Sep. 2025–Jan. 2026
 - **Mathematics Courses:** Topics in Statistics (MCMC), Advanced Operation Research: Stochastic Programming.
 - **Computer Science Courses:** Probability Machine Learning

Peking University

Beijing, China

- *B.S. in Physics, School of Earth and Space Sciences* Sep. 2023–Jun. 2024

RESEARCH EXPERIENCES



Mechanistic Interpretability of Chain-of-Thought Reasoning

University of Wisconsin–Madison

- *Advisor: Prof. Yiqiao Zhong, UW–Madison* Apr. 2026 – Present
 - Designed an interpretability pipeline combining probing, steering, ablation, and activation patching to identify and causally validate internal counting representations and their role in chain-of-thought reasoning.
 - Built synthetic counting tasks with known latent states to study how GPT-2-style Transformers encode running counts and compress sequence history.
 - Designed realistic needle-in-a-haystack prompts and localized the layers and token positions where open-weight LLMs track counts during chain-of-thought reasoning.

Pareto Frontier Model Identification from Pairwise Human Preferences

University of Pennsylvania

- *Advisor: Prof. Zhimei Ren, UPenn* May. 2026 – Present
 - Established theoretical guarantees for Pareto-set identification under Bradley–Terry feedback, leveraging Borda-score estimation to derive minimax sample-complexity bounds.
 - Designed synthetic scaling experiments and benchmarks, developed a tailored maximum-likelihood estimator, and built data-preprocessing and robustness-analysis pipelines for real-world pairwise comparisons.

SELECTED COURSE PROJECTS



LLM Applications: WeChat Content Agent

Peking University

- *Advisor: Prof. Shanghang Zhang, PKU* Apr. 2025 – Jun. 2025
 - Built a Dify-based LLM agent that automated content generation, responsive HTML/CSS typesetting, and draft publishing through Python and WeChat APIs.
 - Integrated RAG and web-retrieval tools to support source-grounded writing and dynamic content collection.

Empirical Asset Pricing and Portfolio Management

Peking University

Mar. 2025 – Jun. 2025

▪ Advisor: Prof. Hai Huang, PKU

- Analyzed Chinese A-share data and constructed dynamically rebalanced portfolios using mean–variance optimization and fundamental signals.
- Evaluated mutual-fund timing and stock-selection performance using the Treynor–Mazuy and Henriksson–Merton models.

[GitHub](#)

Quantitative Trading Backtesting Framework

Peking University

Sep. 2024 – Jan. 2025

▪ Advisor: Dr. Changhao Jiang, PKU

- Developed a modular Python backtesting framework that decoupled signal generation, order execution, and performance evaluation.
- Implemented and benchmarked momentum, mean-reversion, cross-sectional, and time-series strategies using vectorized data processing.

[GitHub](#)

SELECTED HONORS AND AWARDS



Yu Hao Dream Scholarship

Peking University

May. 2026

▪ Scholarship supporting international exchange program

SKILLS & OTHERS



- **Computer Skills:** Python (&PyTorch), R, C++ (Elementary), L^AT_EX
- **Language:** Mandarin Chinese (Native), English (Fluent)
- **The Graduate Record Examinations (GRE):** Total 325 with 155 (Verbal) + 170 (Quant)